**Exercise: The Meeting Report**

**Premises** (translate to symbols first if you like):

Let:

**a** = Ann attends the meeting

**b** = Bob presents

**c** = Carla presents

**r** = The report is ready

If Ann attends the meeting, then Bob or Carla will present.

**a → (b ∨ c)**

If Bob presents, the report will be ready.

**b → r**

If Carla presents, the report will be ready.

**c → r**

The report is not ready.

**¬r**

**Goal:** Prove that Ann did not attend the meeting: **¬a**

**Convert each premise to Conjunctive Normal Form (CNF)** — a conjunction of clauses, each a disjunction of literals. (Remember: X → Y is equivalent to ¬X ∨ Y.)
**Negate the goal** and add it to your set of clauses (this is the refutation setup — you're assuming ¬a is false, i.e., assuming a, and trying to derive a contradiction).
**Apply resolution** repeatedly: from two clauses containing a literal and its negation, derive a new clause containing the remaining literals from both.
**Stop when you derive the empty clause** (□), which signals a contradiction — proving the original goal ¬a must hold.
Write out each resolution step, noting which two clauses (and which literal) you resolved on.

**Step 1 — Convert premises to CNF**
(These are already single clauses, so no further splitting is needed.)

| # | Premise | CNF |
|---|---|---|
| C1 | a → (b ∨ c) | ¬a ∨ b ∨ c |
| C2 | b → r | ¬b ∨ r |
| C3 | c → r | ¬c ∨ r |
| C4 | ¬r | ¬r |

**Step 2 – Negate the goal C5: a**

**Step 3 – Resolution**

| Step | Resolve | On literal | Result |
|---|---|---|---|
| C6 | C2 (¬b ∨ r) + C4 (¬r) | r | ¬b |
| C7 | C3 (¬c ∨ r) + C4 (¬r) | r | ¬c |
| C8 | C1 (¬a ∨ b ∨ c) + C5 (a) | a | b ∨ c |
| C9 | C8 (b ∨ c) + C6 (¬b) | b | c |
| C10 | C9 (c) + C7 (¬c) | c | □ (empty clause) |

**Exercise 2: The Camping Trip**

**Literals:**

**p** = it rains

**q** = we go hiking

**r** = we bring the tent

**s** = we stay at the cabin

**Premises:**

If it does not rain, we go hiking.

**¬p → q**

If we go hiking, we bring the tent.

**q → r**

If we don't bring the tent, we stay at the cabin.

**¬r → s**

We stay at the cabin.

**s**

**Step 1 — Convert premises to CNF**
(These are already single clauses, so no further splitting is needed.)

| # | Premise | CNF |
|---|---|---|
| C1 | ¬p → q | p ∨ q |
| C2 | q → r | ¬q ∨ r |
| C3 | r → ¬s | ¬r ∨ ¬s |
| C4 | s | s |

**Step 2 – Negate the goal C5: ¬p**

**Step 3 – Resolution**

| Step | Resolve | On literal | Result |
|---|---|---|---|
| C6 | C4 (s) + C3 (¬r ∨ ¬s) | s | ¬r |
| C7 | C6 (¬r) + C2 (¬q ∨ r) | r | ¬q |
| C8 | C7 (¬q) + C1 (p ∨ q) | q | p |
| C9 | C8 (p) + C5 (¬p) | p | □ |